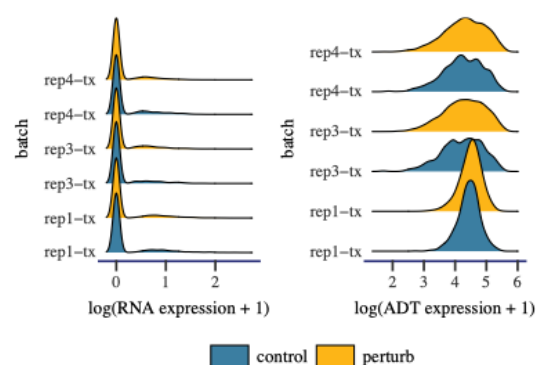
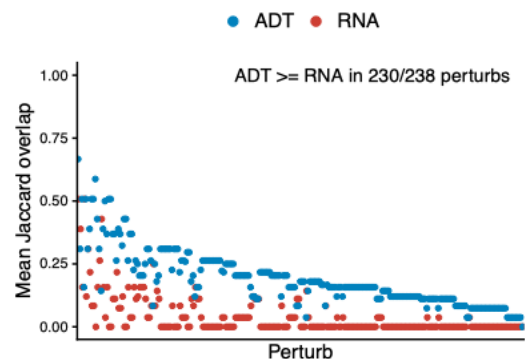


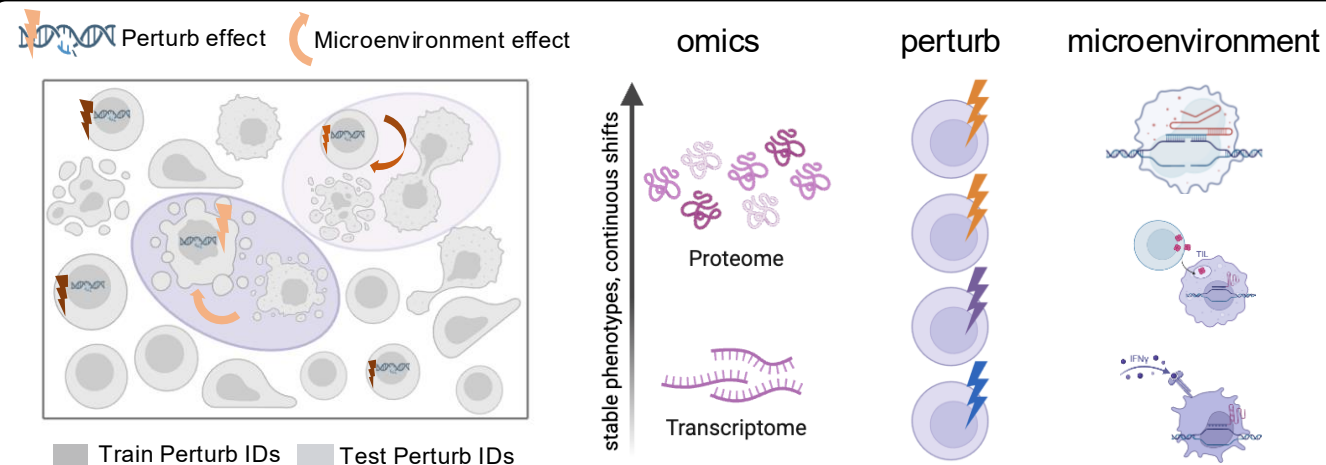
a Motivation: Why multi-omics perturb?

Cross-batch stability of perturb-response

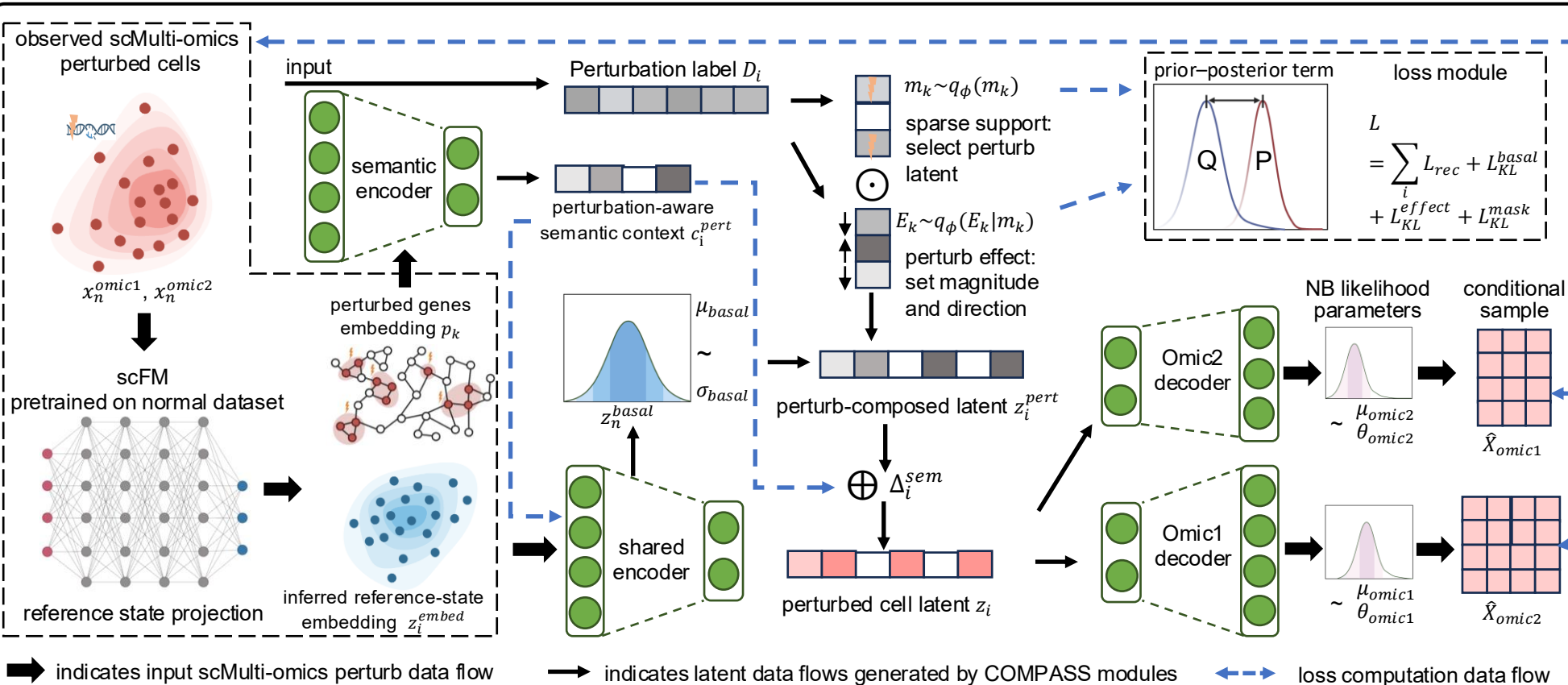
Matched marker consistency across omics

**b**

Input: Single cell multi-omics perturbation data

**c**

COMPASS algorithm

**d**

Downstream analysis

Predicting unseen gene-microenvironment effects to identify immune-state regulators

